

APR-18/BE/Insem.-89

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BE-In Semester
B.E. Computer Engineering
HIGH PERFORMANCE COMPUTING
(2012 Pattern) (410450) (Semester – II)

Time: 1 Hours

Max Marks: 30

Instructions to the candidates:

1. Answer any three questions
2. Neat diagrams must be drawn whenever necessary
3. Assume suitable data if necessary

- Q1. A] Explain N wide Superscalar Architecture in detail [4]
B] Explain Superscalar execution in terms of horizontal waste and vertical waste with example [6]

OR

- Q2. A] Explain following models : [6]
i. MIMD ii. SIMD
B] Explain Multi-Threaded Architecture in detail [4]
Q3. A] Explain Memory Hierarchy and Thread Organization [4]
B] Explain Interconnect, Memory Organization and Programming in details [6]

OR

- Q4. A] Explain Characteristics of Tasks [5]
B] Wire a note on NVIDIA Tesla GPU [5]
Q5. A] Draw and Explain topologies and embeddings [6]
B] Illustrate MPI Routines [4]

OR

- Q6. A] Explain Sending and Receiving Messages using MPI [6]
B] Write syntax and explain the following instruction [4]
i. MP_Barrier ii. MPI_Bcast

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